

Decision Trees

Decision Model

Guess the Animal Game

- ▶ I have an animal in mind
- ▶ You can ask a set of questions.
- ▶ You have to guess (figure out) the animal based on my answers
 - ▶ Of course my answers will be truthful

Analyze the Question!

- ▶ What questions to ask?
- ▶ Why that question?
- ▶ More generally,
 - ▶ What type of questions to ask?
 - ▶ How to decide that?

Analyze this Question!

- ▶ How about asking
“Does it have stripes?”
- ▶ How to decide?
- ▶ Let us look at some data

Guess the Animal – Questions

- ▶ How many legs
- ▶ Does it fly?
- ▶ Is it a wild animal?
- ▶ Is it nocturnal
- ▶ Fur/Feather?
- ▶ Farm Animal?



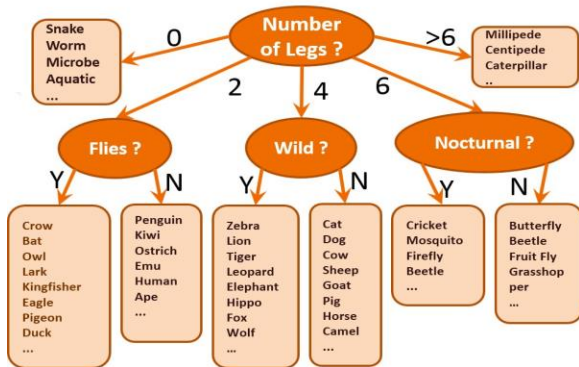
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Aside

- ▶ Why Tree?
 - ▶ CS Term
- ▶ Related terms
 - ▶ Node
 - ▶ Sibling, child, parent
 - ▶ Root, branch, leaf

What is a Good Question?

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- ▶ More precisely, a question that reduces the uncertainty the most
Mathematically, reduces the entropy the most

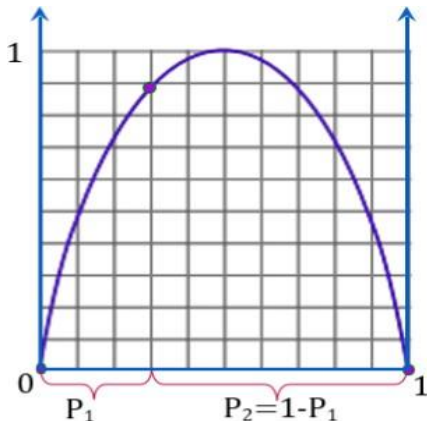
What is Entropy?

- ▶ Measure of **Uncertainty**
 - ▶ For a set containing two classes:

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$$H = -P_1 \log_2 P_1 - P_2 \log_2 P_2$$

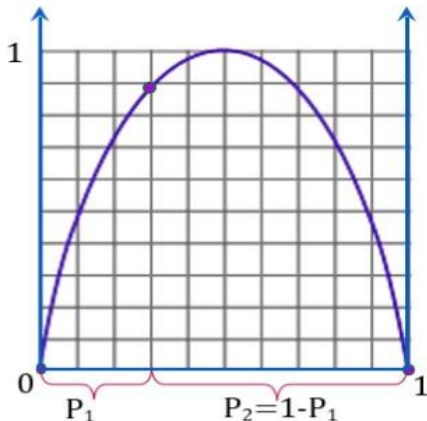


What is Entropy?

- ▶ Measure of **Uncertainty**
 - ▶ For a set containing two classes:

$$H = -P_1 \log_2 P_1 - P_2 \log_2 P_2$$

- ▶ Measure of Impurity
 - ▶ Not the only one



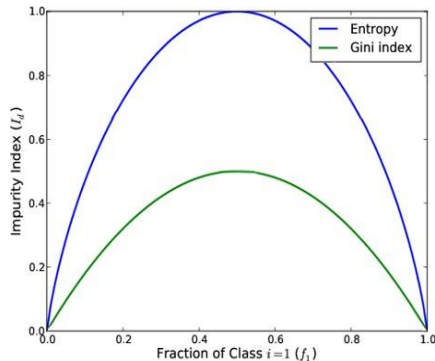
Impurity Metrics

► Entropy:

$$H(x) = - \sum_{i=0}^j P(i) \log_2 P(i)$$

► GINI

$$I(P) = \sum_{i=0}^j P_i(1-P_i) = 1 - \sum_{i=0}^j p_i^2$$

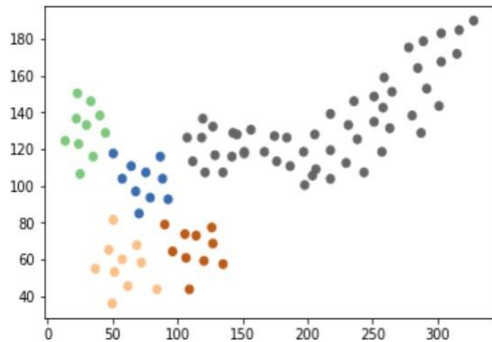


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Experiment

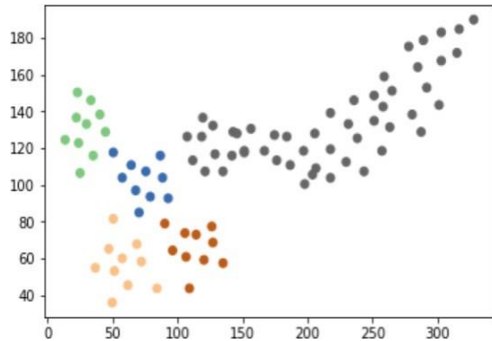
- ▶ Demo_Entropy_Animals
 - ▶ Dataset : Animals
 - ▶ Objective : To understand how to calculate the entropy

Initial Entropy



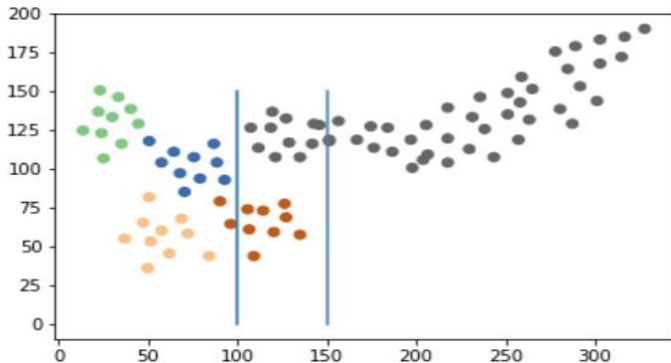
Initial Entropy

$$\begin{aligned}\text{Initial Entropy} &= -\frac{10}{91} \log_2 \left(\frac{10}{91} \right) \\ &\quad -\frac{10}{91} \log_2 \left(\frac{10}{91} \right) \\ &\quad -\frac{10}{91} \log_2 \left(\frac{10}{91} \right) \\ &\quad -\frac{10}{91} \log_2 \left(\frac{10}{91} \right) \\ &\quad -\frac{51}{91} \log_2 \left(\frac{51}{91} \right) \\ &= 1.86\end{aligned}$$



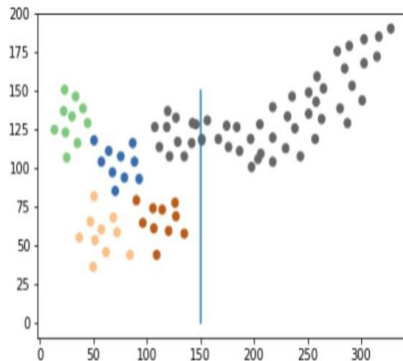
Which split is better?

Which split is better?



Entropy for the split -1

Initial Entropy=1.86



Entropy for the split -1

Initial Entropy=1.86

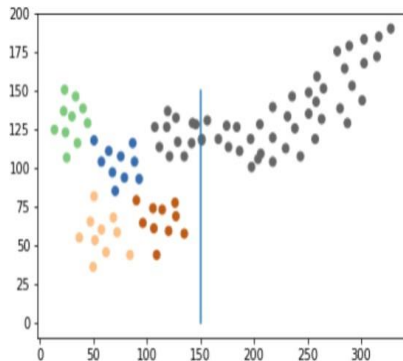
$$H_1 = -\frac{10}{51} \log_2 \left(\frac{10}{51} \right) - \frac{10}{51} \log_2 \left(\frac{10}{51} \right) - \frac{10}{51} \log_2 \left(\frac{10}{51} \right) - \frac{10}{51} \log_2 \left(\frac{10}{51} \right) - \frac{11}{51} \log_2 \left(\frac{11}{51} \right)$$

=2.322(Left side Entropy)

$$H_2 = -\frac{40}{40} \log_2 \left(\frac{40}{40} \right) = -1 \log_2 1$$

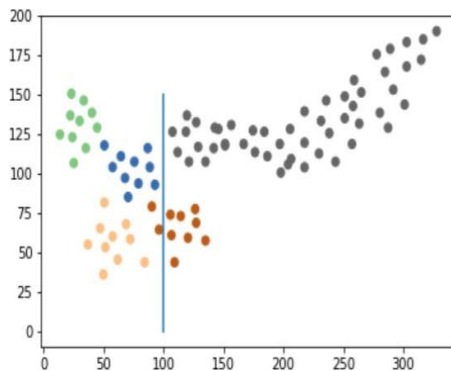
=0(Right side Entropy)

$$\text{Total Entropy} = \left(\frac{51}{91} \right) H_1 + \left(\frac{40}{91} \right) H_2 = 1.301$$



Entropy for the split -2

Initial Entropy=1.86



Entropy for the split -2

Initial Entropy=1.86

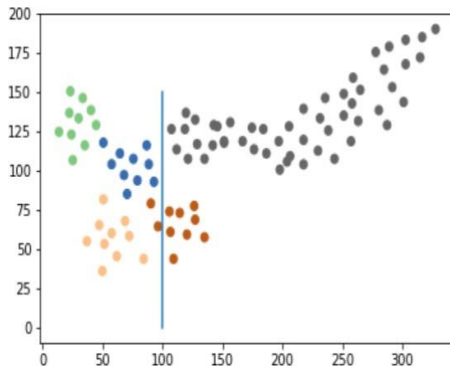
$$H_1 = -\frac{10}{32} \log_2 \left(\frac{10}{32} \right) - \frac{10}{32} \log_2 \left(\frac{10}{32} \right) - \frac{10}{32} \log_2 \left(\frac{10}{32} \right) - \frac{2}{32} \log_2 \left(\frac{2}{32} \right)$$

=1.823(Left side Entropy)

$$H_2 = -\frac{8}{59} \log_2 \left(\frac{8}{59} \right) - \frac{51}{59} \log_2 \left(\frac{51}{59} \right)$$

=0.571(Right side Entropy)

$$\text{Total Entropy} = \left(\frac{32}{91} \right) H_1 + \left(\frac{59}{91} \right) H_2 = 1.011$$



DT in Scikit Learn

Training:

```
from sklearn.tree import DecisionTreeClassifier
clf = DecisionTreeClassifier(criterion = "entropy",
                             max_depth = 5)
clf.fit(x_train, y_train)
Z = clf.predict(x,y)
```

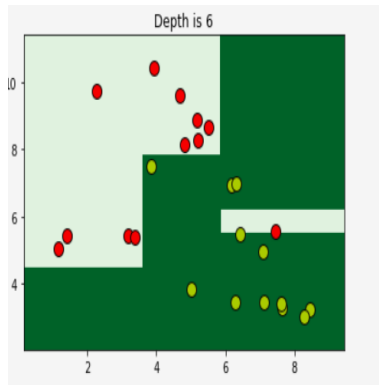
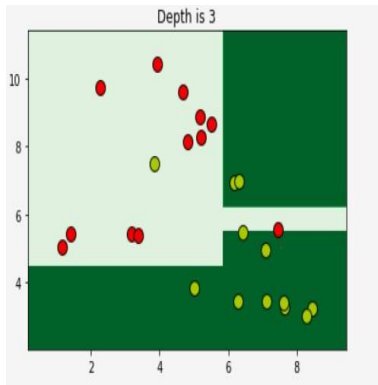
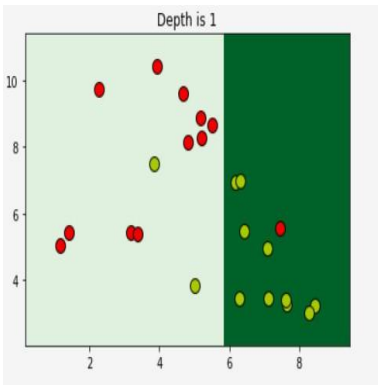
(Dis)advantages of Decision Trees

- ▶ Fast, Compact and Effective
 - ▶ Handles categorical variables *
 - ▶ Interpretable as a set of rules
 - ▶ Can indicate the most useful features
- *Sklearn doesn't support
- ▶ Not suitable for prediction of continuous attribute
 - ▶ Does not handle non-rectangular regions well
 - ▶ Computationally expensive to train
 - ▶ Tends to overfit

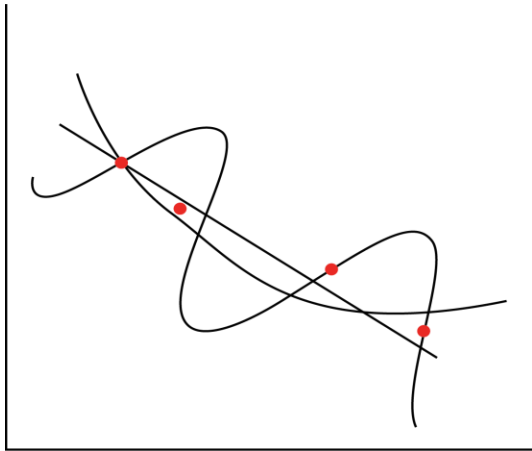
Experiment

- ▶ Demo_DT_Overfitting
 - ▶ Objective : To understand how a decision tree classifier can overfit the data

DT_Overfitting



Overfitting Example



Early Stopping and Pruning

- ▶ Decision trees are notorious for overfitting
- ▶ Early Stopping
 - ▶ Do not split beyond a point
- ▶ Pruning
 - ▶ Once the tree is formed, remove weakest branches
 - ▶ Use validation set to decide when to stop
- ▶ Both approaches also reduce the depth and improve classification speed

Decision Tree Classifier Parameters

- ▶ Early Stopping - **min_impurity_split**
 - ▶ Threshold for early stopping in tree growth
- ▶ Pruning - **min_samples_leaf**
 - ▶ The minimum number of samples required to be at leaf node
 - ▶ This parameter helps limit the growth of the tree

Handling missing values

- ▶ Impute values
 - ▶ Impute 0
 - ▶ Mean/median imputation
 - ▶ Impute from observed values: build predictor
 - ▶ Impute from last observation

Applications of Decision Trees

- ▶ Medical diagnosis
- ▶ Credit risk analysis

Thanks!!
Questions?